

Programming for AI (2+1)

Course Description:

This course provides students with a strong programming foundation for Artificial Intelligence applications. It focuses on writing efficient, well-structured, and well-documented Python code while introducing students to collaborative development practices using version control systems. The course further equips students with hands-on experience in implementing basic Machine Learning models using standard frameworks and tools. By the end of the course, students will be able to develop reproducible AI solutions, manage AI projects collaboratively, and apply machine learning techniques to real-world datasets.

Course Learning Outcomes (CLOs):

By the end of this course, the student will be able to:

CLO-1: Write efficient and well-documented code.
CLO-2: Utilize version control (Git) to manage and collaborate on AI-related code repositories.
CLO-3: Implement basic Machine Learning models using standard ML frameworks.

CLO-wise Assessment and Weightage

CLO	Assessment Tools	Weightage (%)
CLO-1	Quizzes, Programming Assignments, Mid-Term Examination	40%
CLO-2	Git-based Assignment, Project Repository Evaluation	20%
CLO-3	ML Assignments, Project, Final Examination	40%

CLO–GA Mapping with Justification

CLO	Mapped GA	Justification
CLO-1	GA-4: Design / Development of Solutions	This CLO enables students to design, implement, and document computational solutions following accepted programming standards, supporting the development of effective and maintainable software.
CLO-2	GA-6: Individual and Teamwork	This CLO develops students' ability to work effectively as individuals and team members by using version control systems for collaborative software development.
CLO-3	GA-5: Modern Tool Usage	This CLO equips students with hands-on experience in selecting and applying modern machine learning frameworks and tools to solve real-world AI problems.

Week No.	Lab Contents
Week 01	Python environment setup (Python, IDE/Jupyter), basic syntax, print() with sep & end, input/output, variables, data types, simple programs (student info, unit conversion).
Week 02	Conditional statements (if, if-else, nested if), loop constructs (for, while), menu-driven programs, grade calculator, number guessing game.
Week 03	Lists and list operations, slicing, list comprehensions, tuples, dictionaries, dictionary-based student record system, word/number frequency counter.
Week 04	User-defined functions, parameters and return values, lambda functions, map(), filter(), function-based result card generator.
Week 05	File handling (read/write text and CSV files), exception handling using try-except, file-based data storage and retrieval, simple login system.
Week 06	Object-Oriented Programming: classes and objects, constructors, methods, inheritance, encapsulation, OOP-based mini application (AI course or student management system).
Week 07	Introduction to Machine Learning, understanding datasets, basic Pandas operations, reading CSV files, data inspection and preprocessing workflow.
Week 08	Data cleaning techniques (handling missing values, encoding categorical data, normalization), data visualization using Matplotlib, exploratory data analysis.
Week 09	Advanced data visualization using Seaborn: bar plots, box plots, heatmaps, pair plots, correlation analysis, visual data analysis task.
Week 10	Classification models: KNN and Logistic Regression using Scikit-learn, train-test split, model training, evaluation using accuracy, precision, recall.
Week 11	Regression models: Linear and Polynomial Regression, model fitting, prediction, evaluation using MSE and R ² score, regression-based mini task.
Week 12	Decision Trees and Random Forest models, feature importance, model comparison, basic explainability concepts in machine learning.
Week 13	Version control using Git: Git installation, repository creation, commits, push/pull operations, branching, collaborative workflow using GitHub.
Week 14	Deployment of ML models using Streamlit, building a simple ML web app, introduction to web scraping using BeautifulSoup.
Week 15	Advanced web scraping: extracting structured data, storing scraped data in CSV files, cleaning and preparing scraped data for analysis.
Week 16	Python database connectivity (SQLite/MySQL), CRUD operations, integration of database with Python programs, final project demonstration and revision.